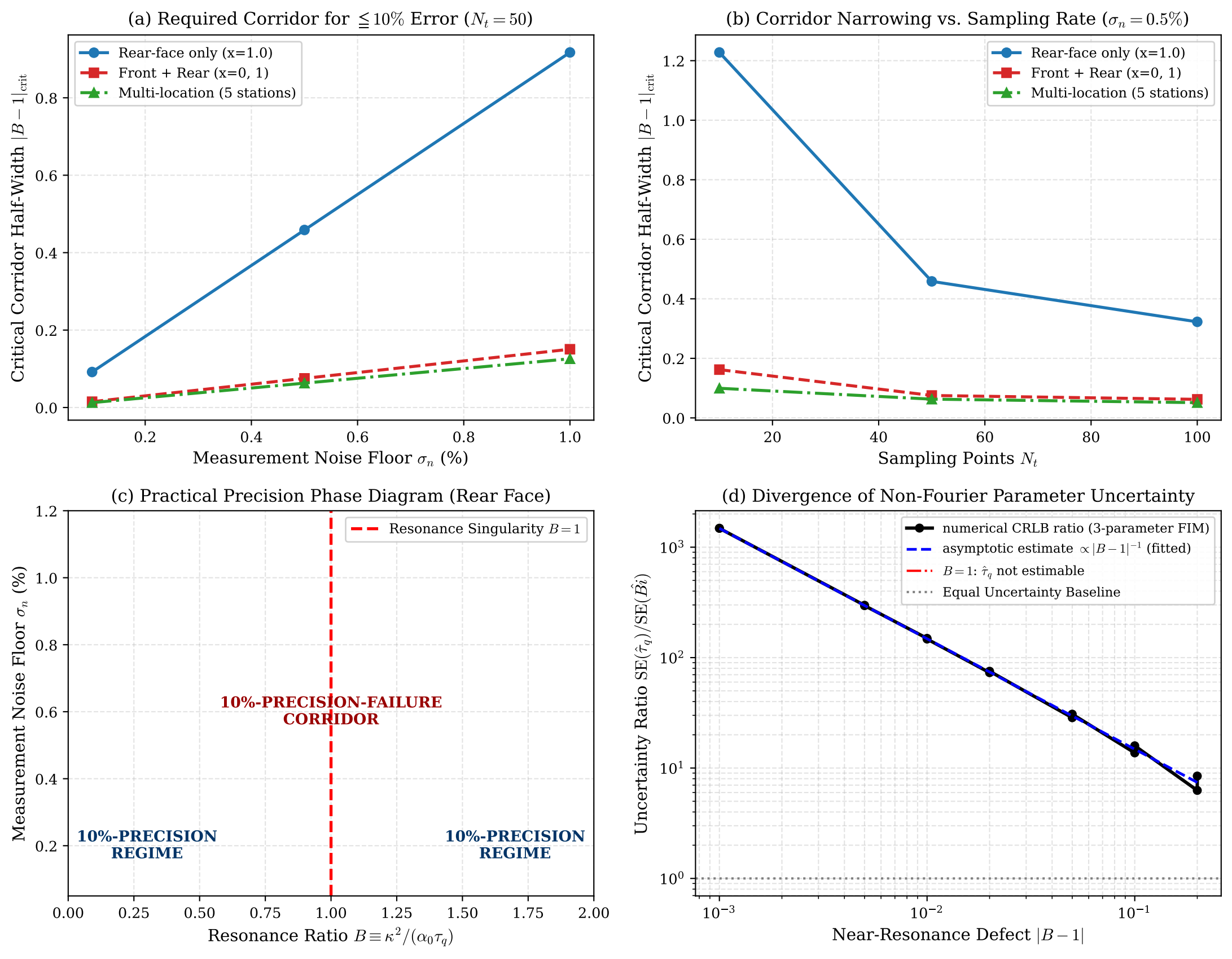

(a) Required Corridor for ≦10% Error ($N_t = 50$)
Rear-face only (x=1.0)
Front + Rear (x=0, 1)
Multi-location (5 stations)
Critical Corridor Half-Width $|B-1|_{\mathrm{crit}}$
Measurement Noise Floor $\sigma_n$ (%)
(b) Corridor Narrowing vs. Sampling Rate ($\sigma_n = 0.5\%$)
Rear-face only (x=1.0)
Front + Rear (x=0, 1)
Multi-location (5 stations)
Critical Corridor Half-Width $|B-1|_{\mathrm{crit}}$
Sampling Points $N_t$
(c) Practical Precision Phase Diagram (Rear Face)
Resonance Singularity $B=1$
10%-PRECISION-FAILURE CORRIDOR
10%-PRECISION REGIME
10%-PRECISION REGIME
Measurement Noise Floor $\sigma_n$ (%)
Resonance Ratio $B \equiv \kappa^2/(\alpha_0 \tau_q)$
(d) Divergence of Non-Fourier Parameter Uncertainty
numerical CRLB ratio (3-parameter FIM)
asymptotic estimate $\propto |B-1|^{-1}$ (fitted)
$B=1$: $\hat{\tau}_q$ not estimable
Equal Uncertainty Baseline
Uncertainty Ratio $\mathrm{SE}(\hat{\tau}_q)/\mathrm{SE}(\hat{Bi})$
Near-Resonance Defect $|B-1|$